

Operating System Security Checklist – Kali Linux

This document demonstrates operating system security fundamentals using Kali Linux. The objective is to understand user privileges, permissions, services, firewall configuration, and operating system hardening in order to reduce the system attack surface.

1. User Accounts and Privileges

Identify the current user:

```
whoami  
kali
```

Display user and group information:

```
id  
uid=1000(kali) gid=1000(kali) groups=1000(kali),27(sudo)
```

2. File Permissions and Ownership

List file permissions:

```
ls -l  
-rw-r--r-- 1 kali kali 4096 report.txt  
-rwxr-xr-- 1 kali kali 2048 script.sh
```

Modify file permissions:

```
chmod 640 report.txt
```

Change file ownership:

```
sudo chown kali:kali report.txt
```

3. Running Processes and Services

View running processes:

```
ps aux | head
```

List running services:

```
systemctl list-units --type=service --state=running  
cron.service  
ssh.service  
NetworkManager.service
```

4. Disabling Unnecessary Services

Disable Bluetooth service:

```
sudo systemctl stop bluetooth  
sudo systemctl disable bluetooth
```

5. Firewall Configuration (UFW)

Enable firewall and check status:

```
sudo ufw enable  
sudo ufw status  
Status: active
```

Set default firewall rules:

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
```

6. Log Monitoring

View authentication logs:

```
cat /var/log/auth.log
sudo: kali : COMMAND=/usr/bin/apt update
```

7. OS Hardening Best Practices

- 1 Use standard user accounts for daily activities
- 2 Apply security updates regularly
- 3 Enable and configure firewall rules
- 4 Disable unnecessary services
- 5 Monitor logs and running processes
- 6 Apply the principle of least privilege

8. Conclusion

Operating system security is a critical foundation of cybersecurity. By applying proper access controls, managing permissions and services, configuring firewall rules, and monitoring logs, the attack surface of Kali Linux can be significantly reduced.